NCEA Level 1 Mathematics and Statistics
91947 · Demonstrate mathematical reasoning
Excellence-level practice questions

SOLUTIONS

About these solutions

- These are full worked solutions to the 91947 practice excellence questions. Where a question can be done more than one way, an alternate solution is also shown.
- They are not an official NZQA assessment schedule, and no marks or grades are given. However, solving these levels of questions in the exam should award you an excellence mark.

How to use them

- Attempt each question properly first, then check your final answer against the answer schedule.
- Only read a solution once you're stuck or have finished, and don't just copy it. Close it and try the question again on your own.
- Your method doesn't have to match the one shown, as long as it is correct and fully explained.

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- If you spot an error or typo, please let me know so I can fix it.

Good luck!

Assessment Schedule – Practice Excellence Questions

Level 1 Mathematics and Statistics: Mathematical Reasoning (91947)

Evidence Statement

Q	Solution
1	Cylinder: $V = \pi r^2 h = \pi(3)^2(8) = 72\pi$ Cone: $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(3)^2(4) = 12\pi$ $V_{\text{total}} = 72\pi + 12\pi = 84\pi \approx 263.894\ \text{m}^3$ Mass $= 263.894 \times 760 = 200\,559$ kg $= 200.559$ tonnes $\dfrac{200.559}{15} = 13.37\ldots$ 13 loads would carry only $13 \times 15 = 195$ tonnes, which is not enough, so **14 truck loads** are needed.
2	(i) The triangle is equilateral, so all three sides are equal. The perimeter is 60 cm, so each side is 20 cm. $2x + y = 20$ (1) $\quad x + 3y = 20$ (2) (1) $\times 3$: $6x + 3y = 60$. Subtracting (2): $5x = 40$, so $x = 8$. From (1): $y = 20 - 16 = 4$. Check CA: $3(8) - 4 = 20$ ✓ $x = 8, \quad y = 4$ (ii) Drop a perpendicular from one vertex to the opposite side. It bisects that side, giving a right-angled triangle with hypotenuse 20 and base 10: $h^2 + 10^2 = 20^2 \Rightarrow h = \sqrt{300} = 10\sqrt{3}$ Area $= \frac{1}{2} \times 20 \times 10\sqrt{3} = 100\sqrt{3} \approx 173\ \text{cm}^2$
3	(i) First differences: $9, 15, 21, 27$. Second differences: $6, 6, 6$. The second differences are constant, so the relationship is quadratic, with $a = \frac{6}{2} = 3$. $y = 3x^2 + bx + c$ $x = 1$: $3 + b + c = 5 \Rightarrow b + c = 2$ $x = 2$: $12 + 2b + c = 14 \Rightarrow 2b + c = 2$ Subtracting: $b = 0$, so $c = 2$. $y = 3x^2 + 2$ (ii) $3x^2 + 2 = 245 \Rightarrow 3x^2 = 243 \Rightarrow x^2 = 81 \Rightarrow \mathbf{x = 9}$ (taking the positive value)

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Q	Solution
4	(i) AC is the diagonal of the base ABCD: $AC^2 = 12^2 + 9^2 = 144 + 81 = 225 \Rightarrow AC = 15$ cm CG is vertical, so triangle ACG is right-angled at C: $AG^2 = 15^2 + 8^2 = 225 + 64 = 289 \Rightarrow AG = \mathbf{17}$ cm (ii) The angle between AG and the base is angle GAC: $\tan\theta = \dfrac{CG}{AC} = \dfrac{8}{15} \Rightarrow \theta \approx \mathbf{28.1}^\circ$
5	The water forms a cone that is similar to the whole cone, so the radii and heights are in the same ratio: $\dfrac{r}{12} = \dfrac{20}{30} \Rightarrow r = 8$ cm $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(8)^2(20) = \dfrac{1280\pi}{3} \approx 1340$ cm^3
6	Perimeter: $2\big[(x-3)+(2x+1)\big] = 2(3x-2) = 6x-4$ $6x - 4 \geq 40 \Rightarrow 6x \geq 44 \Rightarrow x \geq \frac{22}{3} = 7.33\ldots$ Width: $x - 3 < 10 \Rightarrow x < 13$ (The width must also be positive, $x > 3$, which is already satisfied.) $\frac{22}{3} \leq x < 13$ So $x = \mathbf{8}, \mathbf{9}, \mathbf{10}, \mathbf{11}, \mathbf{12}$.
7	(i) Let the three consecutive integers be $n-1$, n and $n+1$. $(n-1) + n + (n+1) = 3n$ $3n$ is 3 times an integer, so the sum is always a multiple of 3. (ii) Let the four consecutive integers be n, $n+1$, $n+2$ and $n+3$. $n + (n+1) + (n+2) + (n+3) = 4n + 6 = 4(n+1) + 2$ $4(n+1)$ is a multiple of 4, so the sum is always exactly 2 more than a multiple of 4. A multiple of 4 leaves a remainder of 0, not 2, so the sum is never a multiple of 4.
8	(i) rate $= \dfrac{125-35}{8-2} = \dfrac{90}{6} = \mathbf{15}$ litres per minute The rate is the gradient of the line: it is the change in volume divided by the change in time. (ii) $V = 15t + c$. Substituting $(2, 35)$: $35 = 15(2) + c \Rightarrow c = 5$. $V = 15t + 5$ At $t = 0$, $V = 5$, so there were already **5 litres** in the tank. (iii) $15t + 5 = 500 \Rightarrow 15t = 495 \Rightarrow t = \mathbf{33}$ minutes

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Q	Solution
9	(i) Let the width be w m, so the length is $(w + 3)$ m. $w(w + 3) = 40 \Rightarrow w^2 + 3w - 40 = 0 \Rightarrow (w + 8)(w - 5) = 0$ $w = -8$ or $w = 5$. A width cannot be negative, so $w = 5$. The garden is **5 m by 8 m**. (ii) The path adds 1 m on each of the four sides, so the outer rectangle is 7 m by 10 m. Area of path $= (7 \times 10) - 40 = 70 - 40 =$ **30** m^2
10	Triangle ADP is right-angled at A, with AD = 8 cm and angle ADP = 32°. AP is opposite the 32° angle and AD is adjacent to it: $\tan 32^\circ = \dfrac{\text{AP}}{8} \Rightarrow \text{AP} = 8 \tan 32^\circ = 5.00$ cm P lies on AB, and AB = 15 cm: PB $= 15 - 5.00 =$ **10.0** cm
11	The toy's outside surface is the curved surface of the cone plus the curved surface of the hemisphere. The flat circle where they join is inside the solid, so it is not counted. Cone $= \pi r L = \pi(5)(13) = 65\pi$ Hemisphere $= \frac{1}{2}\left(4\pi r^2\right) = 2\pi(5)^2 = 50\pi$ Total $= 65\pi + 50\pi =$ **115**$\pi \approx 361$ cm^2
12	(i) Let c be the cost of a coffee and m the cost of a muffin. $3c + 2m = 25.40$ (1) $\quad 5c + 4m = 44.00$ (2) (1) $\times 2$: $6c + 4m = 50.80$. Subtracting (2): $c = 6.80$. From (1): $2m = 25.40 - 3(6.80) = 25.40 - 20.40 = 5.00 \Rightarrow m = 2.50$. A coffee costs **$6.80** and a muffin costs **$2.50**. (ii) $3c + 2m = a$ and $5c + 4m = b$. Multiplying the first by 2: $6c + 4m = 2a$. Subtracting the second: $c =$ **2a − b**

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Q	Solution
13	(i) Each term is found by multiplying the one before it by the same number: $\frac{18}{6} = 3, \qquad \frac{54}{18} = 3, \qquad \frac{162}{54} = 3$ So the relationship is exponential with base 3: $y = k \times 3^x$. $x = 1$: $3k = 6 \Rightarrow k = 2$. $y = 2 \times 3^x$ (ii) $2 \times 3^x = 4374 \Rightarrow 3^x = 2187 = 3^7 \Rightarrow \mathbf{x = 7}$ (iii) Any THREE of: • the y-intercept is $(0, 2)$, since $2 \times 3^0 = 2$; • the graph is always increasing; • y is always positive, so the graph never touches or crosses the x-axis; • $y = 0$ is a horizontal asymptote as $x \to -\infty$; • each time x increases by 1, y is multiplied by 3.
14	(i) Triangle TMA is right-angled at M: $\tan 38^\circ = \frac{\text{TM}}{40} \Rightarrow \text{TM} = 40 \tan 38^\circ \approx \mathbf{31.3}$ m (ii) Triangle TMB is right-angled at M: $\tan 25^\circ = \frac{\text{TM}}{\text{BM}} \Rightarrow \text{BM} = \frac{31.2514}{\tan 25^\circ} \approx \mathbf{67.0}$ m (iii) A is due west of M and B is due south of M, so angle AMB = 90°. By Pythagoras: $\text{AB}^2 = 40^2 + 67.019^2 = 1600 + 4491.5 = 6091.5$ AB $\approx \mathbf{78.0}$ m
15	The wall takes one length, so the fencing makes two widths of x m and one length: length $= 120 - 2x$ $A = x(120 - 2x) = 120x - 2x^2$ This is a quadratic in x with a negative x^2 coefficient, so its graph is a parabola opening downwards and the greatest area is at the vertex. $A = 0$ when $x = 0$ or $x = 60$, and the vertex lies halfway between the roots: $x = \frac{0 + 60}{2} = 30$ Width = 30 m, length = 120 − 60 = 60 m. $A = 30 \times 60 = \mathbf{1800}$ m^2 **Alternate solution:** Complete the square: $A = -2\left(x^2 - 60x\right) = -2\left[(x - 30)^2 - 900\right] = -2(x - 30)^2 + 1800$ The greatest value is 1800, when $x = 30$.

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Q	Solution
16	(i) $\frac{V_{\text{large}}}{V_{\text{small}}} = \frac{675}{200} = \frac{27}{8}$ For similar solids, the volume scale factor is the cube of the length scale factor: $k^3 = \frac{27}{8} \Rightarrow k = \sqrt[3]{\frac{27}{8}} = \frac{3}{2}$ height $= 12 \times \frac{3}{2} =$ **18** cm (ii) The area scale factor is the square of the length scale factor: $\left(\frac{2}{3}\right)^2 = \frac{4}{9}$, so small : large = **4** : **9**
17	(i) Cube every factor inside the bracket: $\left(2x^3y^2\right)^3 = 2^3x^9y^6 = 8x^9y^6$ $\frac{8x^9y^6}{4x^5y} = 2x^{9-5}y^{6-1} =$ **$2x^4y^5$** (ii) Write every term as a power of 2: $4^{x-2} = \left(2^2\right)^{x-2} = 2^{2x-4}$, $8^{x+1} = \left(2^3\right)^{x+1} = 2^{3x+3}$ $2^{3x+1} \times 2^{2x-4} = 2^{5x-3}$ $2^{5x-3} = 2^{3x+3}$ The bases are equal, so the indices are equal: $5x - 3 = 3x + 3 \Rightarrow 2x = 6 \Rightarrow$ **$x = 3$**

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Q	Solution
18	(i) The x-intercepts are -2 and 6, so $y = a(x+2)(x-6)$ Substituting the y-intercept $(0,-12)$: $-12 = a(2)(-6) = -12a \Rightarrow a = 1$ $y = (x+2)(x-6) = \mathbf{x^2 - 4x - 12}$ (ii) A parabola is symmetrical, so the vertex is halfway between the x-intercepts: $x = \dfrac{-2+6}{2} = 2, \qquad y = 4 - 8 - 12 = -16$ The vertex is $(\mathbf{2}, \mathbf{-16})$. (iii) $x^2 - 4x - 12 = 2x - 12$ $x^2 - 6x = 0 \Rightarrow x(x-6) = 0 \Rightarrow x = 0$ or $x = 6$ $x = 0$: $y = -12$. $\quad x = 6$: $y = 0$. The line meets the parabola at $(\mathbf{0}, \mathbf{-12})$ and $(\mathbf{6}, \mathbf{0})$.
19	(i) Cylinder: $V = \pi r^2 \times 3r = 3\pi r^3$ Hemisphere: $V = \dfrac{1}{2} \times \dfrac{4}{3}\pi r^3 = \dfrac{2}{3}\pi r^3$ $V_{\text{total}} = 3\pi r^3 + \frac{2}{3}\pi r^3 = \frac{9}{3}\pi r^3 + \frac{2}{3}\pi r^3 = \dfrac{11}{3}\pi r^3$ ✓ (ii) $\dfrac{11}{3}\pi r^3 = 2000 \Rightarrow r^3 = \dfrac{6000}{11\pi} = 173.62\ldots$ $r = \sqrt[3]{173.62} \approx \mathbf{5.58}$ cm
20	(i) The hypotenuse is the longest side, $(x+5)$. By Pythagoras: $(x-3)^2 + (x+4)^2 = (x+5)^2$ $\left(x^2 - 6x + 9\right) + \left(x^2 + 8x + 16\right) = x^2 + 10x + 25$ $2x^2 + 2x + 25 = x^2 + 10x + 25$ $x^2 - 8x = 0 \Rightarrow x(x-8) = 0$ $x = 0$ would make the side $x - 3$ negative, so $\mathbf{x = 8}$. (ii) The sides are 5, 12 and 13 cm. Area $= \frac{1}{2} \times 5 \times 12 = \mathbf{30}$ cm^2

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Q	Solution
21	(i) Joining Z to every vertex splits the polygon into n identical isosceles triangles, each with two sides of length p and an angle of $\dfrac{360^\circ}{n}$ at Z. Drop a perpendicular from Z to the outside edge of one of these triangles. It bisects both the edge and the angle at Z, making two right-angled triangles with hypotenuse p and an angle of $\dfrac{180^\circ}{n}$ at Z. half the base $= p\sin\left(\dfrac{180^\circ}{n}\right)$, height $= p\cos\left(\dfrac{180^\circ}{n}\right)$ one triangle $= \frac{1}{2}\times 2p\sin\left(\dfrac{180^\circ}{n}\right)\times p\cos\left(\dfrac{180^\circ}{n}\right) = p^2\sin\left(\dfrac{180^\circ}{n}\right)\cos\left(\dfrac{180^\circ}{n}\right)$ There are n of them, so Area $= n\,p^2\sin\left(\dfrac{180^\circ}{n}\right)\cos\left(\dfrac{180^\circ}{n}\right)$ ✓ (ii) $n = 8$, $p = 120$: Area $= 8(120)^2\sin 22.5^\circ\cos 22.5^\circ = 115\,200\times 0.35355 \approx$ **40 700** cm^2 (More precisely 40 729 cm^2, or about 4.07 m^2.)
22	(i) Complete the square: $y = x^2 + kx + 9 = \left(x + \dfrac{k}{2}\right)^2 + 9 - \dfrac{k^2}{4}$ The smallest value of y is $9 - \dfrac{k^2}{4}$. The parabola touches the x-axis at exactly one point when this minimum is 0: $9 - \dfrac{k^2}{4} = 0 \Rightarrow k^2 = 36 \Rightarrow$ **k = 6** or **k = −6** (ii) The parabola stays above the x-axis when the minimum is positive: $9 - \dfrac{k^2}{4} > 0 \Rightarrow k^2 < 36 \Rightarrow$ **−6 < k < 6** **Alternate solution:** Use the discriminant $k^2 - 4(1)(9) = k^2 - 36$: it is 0 when $k = \pm 6$, and negative when $-6 < k < 6$.

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Q	Solution
23	(i) The base is a square of side $2a$, so its diagonal is $\sqrt{(2a)^2 + (2a)^2} = \sqrt{8a^2} = 2a\sqrt{2}$ The apex sits above the centre of the base, which is half a diagonal, $a\sqrt{2}$, from each corner. With h the perpendicular height: $h^2 + \left(a\sqrt{2}\right)^2 = (3a)^2 \Rightarrow h^2 = 9a^2 - 2a^2 = 7a^2$ $h = \mathbf{a\sqrt{7}}$ (ii) The angle θ between a slant edge and the base is in the right-angled triangle with hypotenuse $3a$ and adjacent side $a\sqrt{2}$: $\cos\theta = \dfrac{a\sqrt{2}}{3a} = \dfrac{\sqrt{2}}{3} \Rightarrow \theta \approx \mathbf{61.9}^\circ$ **Alternate solution:** $\tan\theta = \dfrac{a\sqrt{7}}{a\sqrt{2}} = \sqrt{\dfrac{7}{2}} \Rightarrow \theta \approx 61.9^\circ$
24	Whole pyramid: $V = \frac{1}{3} \times$ base area $\times$ height $= \frac{1}{3} \times 10^2 \times 12 = 400$ cm^3 The small pyramid cut from the top is similar to the whole pyramid. It is half the height, so the length scale factor is $\frac{1}{2}$ and the volume scale factor is $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$ $V_{\text{small}} = \frac{1}{8} \times 400 = 50$ cm^3 $V_{\text{frustum}} = 400 - 50 = \mathbf{350}$ cm$^\mathbf{3}$
25	Any two consecutive odd numbers can be written as $2n - 1$ and $2n + 1$, where n is an integer. $(2n + 1)^2 - (2n - 1)^2 = \left(4n^2 + 4n + 1\right) - \left(4n^2 - 4n + 1\right) = 8n$ $8n$ is 8 times an integer, so the difference is always a multiple of 8. **Alternate solution:** Use the difference of two squares: $(2n + 1)^2 - (2n - 1)^2 = \left[(2n + 1) + (2n - 1)\right]\left[(2n + 1) - (2n - 1)\right] = (4n)(2) = 8n$

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Q	Solution
26	(i) $2x - 3y = 12 \quad (1) \qquad 4x + y = 10 \quad (2)$ From (2): $y = 10 - 4x$. Substituting into (1): $2x - 3(10 - 4x) = 12 \Rightarrow 2x - 30 + 12x = 12 \Rightarrow 14x = 42 \Rightarrow x = 3$ $y = 10 - 12 = -2$ P$(3, -2)$ (ii) Find where each line crosses the y-axis by setting $x = 0$: $(1): \ -3y = 12 \Rightarrow y = -4 \qquad (2): \ y = 10$ The two lines meet the y-axis at $(0, -4)$ and $(0, 10)$, so the side along the y-axis has length $10 - (-4) = 14$. The perpendicular height from P to the y-axis is the x-coordinate of P, which is 3. Area $= \frac{1}{2} \times 14 \times 3 =$ **21** units^2
27	(i) The small cone cut from the top is similar to the whole cone. Their heights are $\frac{1}{3}h$ and h, so the length scale factor is $\frac{\frac{1}{3}h}{h} = \frac{1}{3}$ The volume scale factor is the cube of the length scale factor: $\left(\frac{1}{3}\right)^3 = \frac{1}{27} \quad \checkmark$ (ii) The frustum is what is left: $1 - \frac{1}{27} = \frac{26}{27}$ frustum : small cone $= \frac{26}{27} : \frac{1}{27} =$ **26** : **1**
28	(i) Speed is the gradient of a distance–time graph. Converting each stage to km h^{-1}: Stage 1: 8 km in 20 minutes $= \frac{8}{20/60} =$ **24** km h^{-1} Stage 2: the graph is horizontal, so the gradient is 0: the cyclist is at rest, **0** km h^{-1} Stage 3: $23 - 8 = 15$ km in 20 minutes $= \frac{15}{20/60} =$ **45** km h^{-1} The steeper the section, the greater the gradient and so the faster the cyclist is travelling. (ii) average speed $= \frac{\text{total distance}}{\text{total time}} = \frac{23}{50/60} =$ **27.6** km h^{-1}

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Q	Solution
29	(i) Each pattern is a rectangle: Pattern 1 is 1×4, Pattern 2 is 2×5, Pattern 3 is 3×6. Pattern n is n by $(n+3)$, so $T = n(n+3) = \mathbf{n^2 + 3n}$ **Alternate solution:** Use differences. First differences: $6, 8, 10$; second differences: $2, 2$. These are constant, so T is quadratic with $a = \frac{2}{2} = 1$: $T = n^2 + bn + c$ $n = 1$: $1 + b + c = 4$; $n = 2$: $4 + 2b + c = 10$. Subtracting gives $b = 3$, so $c = 0$. (ii) $n^2 + 3n = 460 \Rightarrow n^2 + 3n - 460 = 0 \Rightarrow (n+23)(n-20) = 0$ $n = -23$ or $n = 20$. A pattern number must be positive, so it is **Pattern 20**. (Check: $20 \times 23 = 460$ ✓)
30	(i) First find the diagonal of the floor, using the 6 m and 4 m sides: $d^2 = 6^2 + 4^2 = 36 + 16 = 52$ The height is perpendicular to the floor, so $\text{AG}^2 = 52 + 3^2 = 61 \Rightarrow \text{AG} = \mathbf{\sqrt{61}} \approx 7.81$ m (ii) The angle between AG and the floor is in the right-angled triangle with opposite side 3 (the height) and adjacent side $\sqrt{52}$ (the floor diagonal): $\tan\theta = \dfrac{3}{\sqrt{52}} \Rightarrow \theta \approx \mathbf{22.6}^\circ$
31	$5(x-2) \geq 2x + 11 \Rightarrow 5x - 10 \geq 2x + 11 \Rightarrow 3x \geq 21 \Rightarrow x \geq 7$ $3 - 2x > -17 \Rightarrow -2x > -20 \Rightarrow x < 10$ (Dividing by -2 reverses the inequality sign.) Both must hold, so $\mathbf{7 \leq x < 10}$ The integer values are $\mathbf{7}, \mathbf{8}, \mathbf{9}$.

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Q	Solution
32	(i) Cylinder surface area $= 2\pi r^2 + 2\pi rh$, Sphere surface area $= 4\pi r^2$ $2\pi r^2 + 2\pi rh = 4\pi r^2$ $2\pi rh = 2\pi r^2$ Dividing both sides by $2\pi r$ (which is not zero): $h = r$ ✓ (ii) $V_{\text{cylinder}} = \pi r^2 h = \pi r^2(r) = \pi r^3$, $V_{\text{sphere}} = \frac{4}{3}\pi r^3$ $V_{\text{cylinder}} : V_{\text{sphere}} = \pi r^3 : \frac{4}{3}\pi r^3 = 1 : \frac{4}{3} = \mathbf{3} : \mathbf{4}$
33	(i) The ladder is the hypotenuse, and 1.2 m is adjacent to the 70° angle: $\cos 70^\circ = \dfrac{1.2}{L} \Rightarrow L = \dfrac{1.2}{\cos 70^\circ} \approx \mathbf{3.51}$ m height $= L \sin 70^\circ = 3.5086 \times \sin 70^\circ \approx \mathbf{3.30}$ m (ii) The ladder keeps the same length, so at 55°: new distance from wall $= L \cos 55^\circ = 3.5086 \times \cos 55^\circ \approx 2.01$ m $2.01 - 1.2 = \mathbf{0.81}$ m further out new height $= L \sin 55^\circ = 3.5086 \times \sin 55^\circ \approx 2.87$ m $3.30 - 2.87 = \mathbf{0.42}$ m down the wall
34	(i) First differences: 4, 8, 16. These are not constant, and each is double the one before, so the relationship is exponential with base 2, plus a constant: $y = A \times 2^x + C$ $x = 0$: $A + C = 7$ (since $2^0 = 1$) $x = 1$: $2A + C = 11$ Subtracting: $A = 4$, so $C = 3$. $y = 4 \times 2^x + 3$ Check $x = 2$: $4(4) + 3 = 19$ ✓ Check $x = 3$: $4(8) + 3 = 35$ ✓ (ii) Any THREE of: • the y-intercept is $(0, 7)$; • the graph is always increasing; • $y = 3$ is a horizontal asymptote as $x \to -\infty$; • y is always greater than 3, so the graph never crosses the x-axis; • each time x increases by 1, the amount above 3 doubles.

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Q	Solution
35	Each ball has radius r and fits tightly, so the tube has radius r. There are n balls stacked, each of diameter $2r$, so the tube's height is $2rn$. $V_{\text{tube}} = \pi r^2(2rn) = 2\pi nr^3$ $V_{\text{balls}} = n \times \frac{4}{3}\pi r^3 = \frac{4}{3}\pi nr^3$ $\frac{V_{\text{balls}}}{V_{\text{tube}}} = \frac{\frac{4}{3}\pi nr^3}{2\pi nr^3} = \frac{4}{3} \times \frac{1}{2} = \frac{2}{3}$ Both r and n cancel, so the balls always fill exactly $\frac{2}{3}$ of the tube, whatever their size and however many there are. empty space $= 1 - \frac{2}{3} = \frac{1}{3} = \mathbf{33\frac{1}{3}}\%$
36	(i) DE is parallel to BC, so angle ADE = angle ABC and angle AED = angle ACB (corresponding angles). Angle A is shared, so triangle ADE is similar to triangle ABC. AB = AD + DB = 6 + 4 = 10 cm scale factor $= \frac{\text{AD}}{\text{AB}} = \frac{6}{10} = \frac{3}{5}$ $\frac{\text{DE}}{\text{BC}} = \frac{3}{5} \Rightarrow \text{BC} = 9 \times \frac{5}{3} = \mathbf{15}$ cm (ii) The area scale factor is the square of the length scale factor: $\frac{\text{area ADE}}{\text{area ABC}} = \left(\frac{3}{5}\right)^2 = \frac{9}{25}$ The trapezium is the rest of triangle ABC: $\frac{\text{area DBCE}}{\text{area ABC}} = 1 - \frac{9}{25} = \frac{16}{25}$ area ADE : area DBCE = **9** : **16**
37	(i) Set the two equations equal to find where they meet: $x^2 + 5x + 10 = 3x + k$ $x^2 + 2x + (10 - k) = 0$ A tangent touches at exactly one point, so this quadratic has exactly one solution and its discriminant is 0: $2^2 - 4(1)(10 - k) = 0 \Rightarrow 4 - 40 + 4k = 0 \Rightarrow 4k = 36 \Rightarrow \mathbf{k = 9}$ **Alternate solution:** Complete the square: $x^2 + 2x + (10 - k) = (x + 1)^2 + 9 - k$. For exactly one solution, $9 - k = 0$, so $k = 9$. (ii) With $k = 9$: $x^2 + 2x + 1 = 0 \Rightarrow (x + 1)^2 = 0 \Rightarrow x = -1$ $y = 3(-1) + 9 = 6$ The line touches the parabola at $(\mathbf{-1, 6})$.

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Q	Solution
38	Let v be the number of vans and b the number of buses. $12v + 40b \geq 200$ (students) $\quad v + b \leq 8$ (vehicles) Cost $= 90v + 250b$ Test each possible number of buses, using the smallest number of vans that still carries 200 students: $b = 6$: $v = 0$ 6 vehicles, 240 students \$1500 $b = 5$: $v = 0$ 5 vehicles, 200 students \$1250 $b = 4$: $v = 4$ 8 vehicles, 208 students \$1360 $b = 3$: $v = 7$ 10 vehicles — too many not allowed With 3 or fewer buses, at least 7 vans are needed, which breaks the 8-vehicle limit. The cheapest allowed combination is **5 buses and no vans**, costing **\$1250**.
39	(i) The small triangle sitting above the square is similar to the whole triangle, because the top of the square is parallel to the side of length b, so the angles match. The whole triangle has height a and base b. The small triangle has height $a - s$ and base s. $\dfrac{a-s}{s} = \dfrac{a}{b}$ $b(a - s) = as$ $ab - bs = as$ $ab = as + bs = s(a + b)$ $s = \dfrac{ab}{a+b}$ $\quad\checkmark$ (ii) $s = \dfrac{6 \times 12}{6 + 12} = \dfrac{72}{18} = \mathbf{4}$ cm
40	The cross-section is an isosceles triangle with base $2 \times 6 = 12$ cm and height 8 cm. Each sloping side is $\sqrt{6^2 + 8^2} = \sqrt{100} = 10$ cm The sphere becomes the circle that touches all three sides, so its radius r satisfies Area $= r \times s$, where s is half the perimeter Area $= \frac{1}{2} \times 12 \times 8 = 48$ cm^2, $\quad s = \dfrac{12 + 10 + 10}{2} = 16$ cm $48 = 16r \Rightarrow r = \mathbf{3}$ cm **Alternate solution:** Use similar triangles. Put the centre on the axis, r above the base, so it is $8 - r$ below the apex. The radius drawn to the point where the circle touches a sloping side is perpendicular to that side, making a right-angled triangle similar to the one with sides 6, 8 and 10: $\dfrac{r}{8-r} = \dfrac{6}{10} \Rightarrow 10r = 48 - 6r \Rightarrow 16r = 48 \Rightarrow r = 3$ cm

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